

Initial Post (Peer A)

by Peer A - Sunday, 3 May 2026, 3:17 PM

Hancock et al. (2024) argue that data and digital technologies can strengthen human rights investigations by expanding access to evidence and making less visible harms easier to identify. Their study, which focuses on maritime human rights concerns, also shows that these tools can intensify risks to fairness and accountability when treated as neutral or purely technical solutions. In human rights investigations, the issue is not only whether data can be collected, but whose experiences are represented, which victims or witnesses remain invisible, and who has authority to interpret evidence.

One fairness risk is that datasets may be incomplete, unevenly produced or shaped by existing inequalities. Abuses affecting marginalised communities may be under-reported because victims fear retaliation, lack digital access or distrust official institutions. If investigators rely heavily on such datasets, the investigation may reproduce existing silences rather than expose harm. This reflects Barocas and Selbst's (2016) argument that data-driven systems can produce discriminatory outcomes even without intentional discrimination, because historical inequalities may already be embedded in the data. As a result, technology may give partial evidence the appearance of objectivity.

Accountability may also be weakened when digital systems make responsibility difficult to trace. If an algorithmic tool identifies a vessel, person or pattern as suspicious, it may be unclear whether an error should be attributed to the developer, the organisation using the system or the investigator relying on its output. This matters because errors may affect victims, accused individuals, witnesses and the credibility of the investigation. Dubberley, Koenig and Murray (2020) emphasise that digital evidence can support human rights documentation only when it is properly verified, contextualised and preserved. Without transparent methods, audit trails and human review, digital evidence may be misinterpreted or over-relied upon.

A further concern arises when technologies are developed without direct input from the communities they are intended to protect. Hancock et al. (2024) suggest that some technological interventions are detached from affected people's lived experiences. This can result in systems that are inaccessible, unsafe, culturally inappropriate or misaligned with local priorities. A reporting platform may increase surveillance or retaliation risks if designers do not understand local power relations. It may also collect sensitive information without meaningful consent, undermining dignity, privacy and autonomy.

Mitigation should begin during design rather than after deployment. Friedman and Hendry's (2019) value sensitive design approach requires designers to consider human values, stakeholders and potential harms throughout development. However, this should be extended through Costanza-Chock's (2020) design justice

perspective, which argues that affected communities should actively shape design decisions, not merely be consulted. In practice, this requires participatory risk assessment, privacy-by-design, accessible interfaces, community testing and clear accountability mechanisms. Technology can support human rights investigations, but it should not replace the voices, safety or agency of the people whose rights are at stake.

References

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Peer Response

by Peer B - Tuesday, 5 May 2026, 12:03 PM

Hello Peer A,

I appreciate your contribution.

Your discussion offers a nuanced analysis of the ways in which data and technology can both support and threaten fairness and accountability in human rights investigations. You mentioned incomplete and unequal datasets, to which an added mitigation strategy would be the use of data quality assessment and bias detection techniques prior to analysis. Data-driven systems ma

You perpetuate existing inequalities if the underlying data is not critically examined (O'Neil, 2016). Therefore, the application of systematic validation procedures can assist in avoiding wrong conclusions resulting from incomplete or biased data.

Your interest in accountability issues in algorithmic systems is also particularly relevant. To address this, organisations should put in place clear governance frameworks and audit trails, so accountability for decisions remains traceable. Floridi et al. (2018) discuss that the design of AI systems should integrate transparency and accountability by means of documentation, explainability, and human oversight. This is in line with your observation that digital evidence has to be carefully verified and contextualised (Dubberley et al., 2020).

Your mention of design justice as a way of addressing community exclusion is very relevant. To go further, participatory approaches should not only be consultations but also empower communities in decision-making processes. This aligns with the principles of inclusive and ethical design that seek to redistribute power in the creation of technology (Costanza-Chock, 2020). Moreover, human rights impact assessments at the design stage are a useful tool for proactively identifying risks related to privacy, consent and potential harm (UNGP, 2011).

To conclude, the risks you have identified can be significantly mitigated by a combination of strong data validation, transparent governance and meaningful community involvement.

Reference

Costanza-Chock, S. (2020) Design Justice: Community-Led Practices to Build the Worlds We Need. Cambridge, MA: MIT Press.

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My Peer Response

by Séba Daher - Monday, 11 May 2026, 5:13 AM

(Evidence: <https://www.my-course.co.uk/mod/forum/discuss.php?d=364101#p736146>)

Hi Peer A and Peer B,

Thank you both for sharing your perspectives. Respectively, your post and response have addressed data bias, audit trails, and participatory design effectively. A dimension that remains underexplored is the direct physical and personal risk that

data and technology can impose on the very communities they are intended to protect, particularly through surveillance and retaliation.

Renn et al. (2021) observe that digitalisation functions as a tool of oppression as readily as liberation, noting that autocratic actors have exploited digital infrastructure to monitor and control dissent. In human rights investigations, this risk is acute: data collection about victims, witnesses, or community informants creates records that, if compromised, could expose individuals to retaliation by the same actors whose conduct is under investigation. The risk is not hypothetical; it materialises whenever data is stored without adequate access controls, transmitted over insecure channels, or held by organisations operating in jurisdictions where legal compulsion to disclose is plausible.

Mitigating this requires a threat-modelling approach applied before deployment, not after harm has occurred. NIST (2002) describes a systematic methodology in which the characteristics of realistic adversaries, including their resources, access, and objectives, are mapped against system vulnerabilities before controls are selected. Applied here, this would mean human rights organisations conducting adversary-aware risk assessments that explicitly consider state-level threat actors, not only internal data governance failures.

This shifts the design conversation: community involvement remains essential, but so does technical adversarial thinking that communities themselves may lack the specialised knowledge to perform independently.

References

NIST (2002) *Risk management guide for information technology systems*. SP 800-30. Gaithersburg, MD: National Institute of Standards and Technology.

Renn, O., Beier, G. and Schweizer, P.-J. (2021) 'The opportunities and risks of digitalisation for sustainable development: a systemic perspective', *GAIA*, 30(1), pp. 23-28.

Initial Post (Peer B)

by Peer B - Tuesday, 5 May 2026, 11:03 AM

Good morning,

The use of data and technology in human rights investigations can enhance efficiency but also generate serious risks for fairness and accountability. Hancock et al. (2024) argue that technological tools such as artificial intelligence (AI), big data analytics, and digital surveillance can embed biases, leading to unfair outcomes. For example, algorithmic bias can emerge when datasets are incomplete, unrepresentative or historically biased, which can perpetuate discrimination against vulnerable groups (O'Neil, 2016). This undermines fairness because decisions may impact populations unequally without transparent justification.

The “black box” aspect of many AI systems, which do not have easily explainable decision processes (Burrell, 2016), may result in further weakening of accountability. This lack of transparency makes auditing or contesting outcomes difficult, a key issue in human rights contexts where decisions need to be justifiable and legally defensible.

There is also the risk that technological solutions may be created without meaningful input from the communities they are trying to protect. Such approaches can create systems that do not reflect local contexts, cultural sensitivities, or real-world needs, thereby compounding harm or excluding key stakeholders (Eubanks, 2018). To reduce these risks, participatory design approaches should be used that engage the affected communities throughout the development process. Furthermore, fairness-aware algorithms, transparency mechanisms, and ethical oversight frameworks should be established to reduce bias and ensure accountability.

Reference

- Burrell, J. (2016) 'How the machine “thinks”: Understanding opacity in machine learning algorithms', *Big Data & Society*, 3(1).
- Eubanks, V. (2018) *Automating Inequality*. New York: St. Martin's Press.
- Hancock, J., Hui, R., Singh, J. and Mazumder, A. (2024) *(Use your exact source title if provided)*
- O'Neil, C. (2016) *Weapons of Math Destruction*. New York: Crown.

Peer Response

by Peer A - Saturday, 9 May 2026, 9:05 AM

Dear Peer B,

Thank you for the clear and well-structured contribution. The post effectively identifies algorithmic bias, the opacity of AI systems and limited community participation as key risks affecting fairness and accountability in human rights investigations. These points are relevant because technological outputs may appear neutral, while still reflecting incomplete datasets, historical discrimination or assumptions embedded during system design.

A useful preventative measure would be to introduce a formal human rights impact assessment before deploying technological tools in investigative settings. This would help assess whether a system could disproportionately affect vulnerable groups, create unjustified surveillance risks or produce decisions that are difficult to challenge. Human rights impact assessment can support accountability by identifying potential harms before systems are implemented (Götzmann, 2017). Hancock et al. (2024) also demonstrate that digital technologies can support human rights investigations, but they may create accountability problems when data is incomplete, biased or interpreted without sufficient context.

Furthermore, transparency should be treated as a design requirement rather than an optional feature. Burrell (2016) argues that algorithmic opacity can make machine learning outcomes difficult to audit. Therefore, clear documentation of data sources, explainable AI methods and audit trails should be included to support review, correction and accountability.

The point on community involvement is also important. Eubanks (2018) shows that automated systems can reinforce harm when they are developed without sufficient understanding of affected communities. This risk could be reduced through participatory design, where communities are involved during problem definition, testing and evaluation. Therefore, ethical oversight, transparent design and meaningful participation should work together to reduce the fairness and accountability risks identified in the post.

References

Burrell, J. (2016) 'How the machine "thinks": Understanding opacity in machine learning algorithms', *Big Data & Society*, 3(1).

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My Peer Response

by Séba Daher - Monday, 11 May 2026, 5:28 AM

(Evidence: <https://www.my-course.co.uk/mod/forum/discuss.php?d=364671#p736149>)

Hi Peer B and Peer A,

Thank you both for your contributions. Respectively, your post and response have done well to identify algorithmic bias, black-box opacity, and the importance of pre-deployment human rights impact assessments. The surveillance and retaliation dimension was explored in my response to Peer A's post, so rather than revisit that ground, I want to build on a specific concern "Peer B" raises: the legal defensibility of technologically derived evidence.

Peer B correctly notes that decisions in human rights contexts must be justifiable and legally defensible. This requirement exposes a deeper problem beyond opacity alone, namely investigator over-reliance. When human investigators defer habitually to algorithmic outputs without the technical literacy to interrogate or challenge them, institutional judgement gradually atrophies. Dubberley et al. (2020) argue that digital evidence supports human rights documentation only when it is properly verified and contextualised. Yet verification presupposes that investigators retain the capability and confidence to question what a system produces, which becomes harder when organisations have structured their workflows around accepting automated outputs as authoritative.

AIRMIC (2010), drawing on ISO 31000, emphasise that embedding risk awareness at the operational level, not only among technical specialists, is a precondition for meaningful accountability. Applied here, this would require organisations to invest in technical literacy training alongside the deployment of any algorithmic tool, ensuring that human oversight remains substantive rather than nominal.

References

AIRMIC (2010) *A structured approach to enterprise risk management (ERM) and the requirements of ISO 31000*. London: AIRMIC.

Dubberley, S., Koenig, A. and Murray, D. (eds.) (2020) *Digital witness: Using open source information for human rights investigation, documentation, and accountability*. Oxford: Oxford University Press.